Chapter 10: One- and Two-Sample Tests of Hypotheses (Part IV)

ENGR 0021

Statistical Significance vs. Practical Significance

- If a result is **statistically significant**, it is hard to explain as chance variation alone.
- But **statistically significant** does **not** necessarily mean **important**.
- These are two different ideas:
  - **Statistical significance:** Is the observed difference likely due to chance?
  - **Practical significance:** Is the difference large enough to matter in the real world?
- A very small difference can be statistically significant if the sample size is large enough.
- An important difference may fail to be statistically significant if the sample size is too small.

Example: Big-City vs. Rural Children

- Investigators compare WISC vocabulary scores for children ages 6–9.
- Two independent simple random samples:
  - Big-city children: $n = 2500$, mean $= 26$, SD $= 10$
  - Rural children: $n = 2500$, mean $= 25$, SD $= 10$
- Difference in sample means:

$$26 - 25 = 1$$

- Estimated standard error:

$$SE(\bar{X}_1 - \bar{X}_2) = \sqrt{\frac{SD_1^2}{n_1} + \frac{SD_2^2}{n_2}}$$

$$SE \approx 0.3$$

- Two-sample z-statistic:

$$z \approx \frac{1}{0.3} \approx 3.3$$

- The P-value is very small, so the difference is **statistically significant**.

Why Statistical Significance Is Not Enough

- The z-test only says the 1-point difference is hard to explain by chance.
- It does **not** say the difference is important.
- On the WISC vocabulary test:
  - There are 40 words
  - A 1-point difference is only about a partial understanding of one word.
- So this result is **highly significant statistically**, but may be **minor in practical terms**.
- Key lesson:
  - Large samples make the standard error small
  - Small differences can then produce large z-values
  - Statistical significance does not automatically imply practical importance

Takeaway

- The P-value tells us whether a result is hard to explain by chance.
- The P-value does **not** tell us whether the result matters.
- Always consider:
  - the size of the effect
  - the context
  - the practical consequences
- Conclusion:

$$\text{Statistical significance} \neq \text{practical significance}$$

Which Statistical Test Should Be Used?

1. Demographers think that about 55% of newborns are male. In a hospital, 568 out of 1,000 births are male. (A), (C)
2. To test whether a die is fair, someone rolls it 600 times. Outcomes are grouped as:
   - even/odd
   - large (4,5,6) / small (1,2,3) (G)
3. Income study:
   - High school: $n = 250$, mean = \$40,000, SD = \$25,000
   - College: $n = 250$, mean = \$70,000, SD = \$30,000

   (B), (D)

   Question: Is the difference in averages real or due to chance?

Choices:

(A) One-sample z-test

(B) Two-sample z-test

(C) One-sample t-test

(D) Two-sample t-test

(E) Chi-square variance test

(F) F variance proportion test

(G) Chi-square goodness-of-fit test

(H) Chi-square test of independence

True or False?

(a) A difference which is highly significant must be very important. False

(b) Big samples are bad because small differences will look significant.

False

Common Misinterpretations of p-values

- Not $P(H_0 \mid \text{data})$:
  - p-value $= P(\text{data as extreme} \mid H_0)$
  - Measures compatibility of data with H_0
- Not "probability due to chance":
  - Computed assuming H_0 is true
- 0.05 is arbitrary:
  - Threshold is a convention, not a scientific boundary
- Does not measure effect size:
  - Small p-value $\neq$ important effect
  - Large n can make trivial effects significant